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Harada

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(54) **DISTAL-END-CAP DETACHMENT JIG**

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Aug. 26, 2020 (JP) 2020-142536

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A61B 1/018 (2006.01)

(52) **U.S. Cl.**
CPC ***A61B 1/00101*** (2013.01); ***A61B 1/00098*** (2013.01); ***A61B 1/00137*** (2013.01); ***A61B 1/018*** (2013.01)

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CPC A61B 1/00101; A61B 1/00098; A61B 1/00137; A61B 1/018; A61B 1/0008; A61B 1/00128; A61B 1/0676; A61M 5/3204
See application file for complete search history.

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(57) **ABSTRACT**

Provided is a distal-end-cap detachment jig capable of safely detaching a distal end cap from a distal-end-portion body. A distal-end-cap detachment jig that is used to detach a distal end cap mounted on a distal end of a side-viewing scope and having an inner space which communicates with a cap opening allowing a passage of a treatment tool, includes at least a body portion that has an inclined surface, in which the inclined surface is tapered in a direction of insertion into the inner space of the distal end cap, and in a case where the body portion is inserted into the inner space of the distal end cap, the inclined surface deforms the distal end cap in a direction of expanding the inner space of the distal end cap.

7 Claims, 10 Drawing Sheets

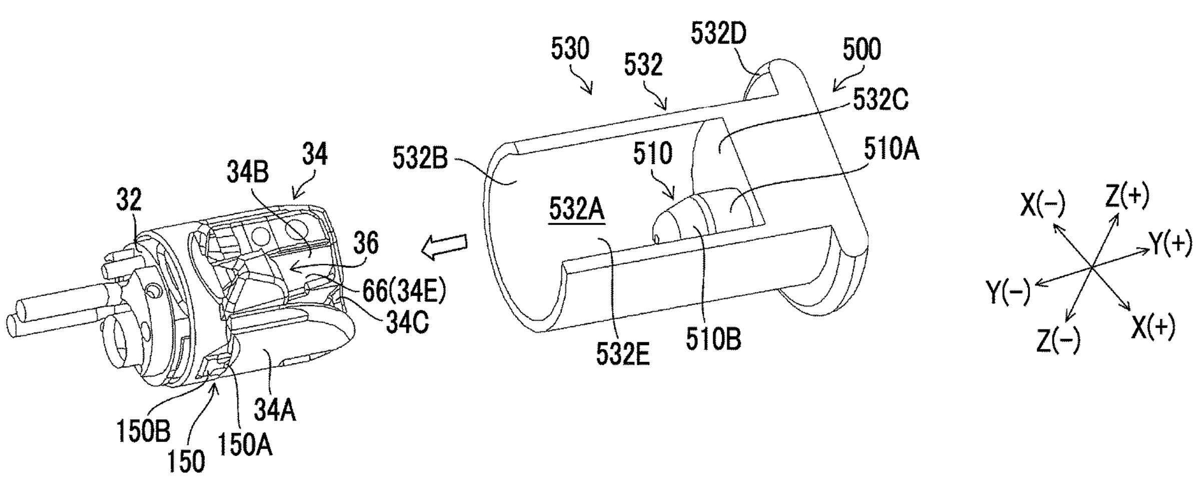
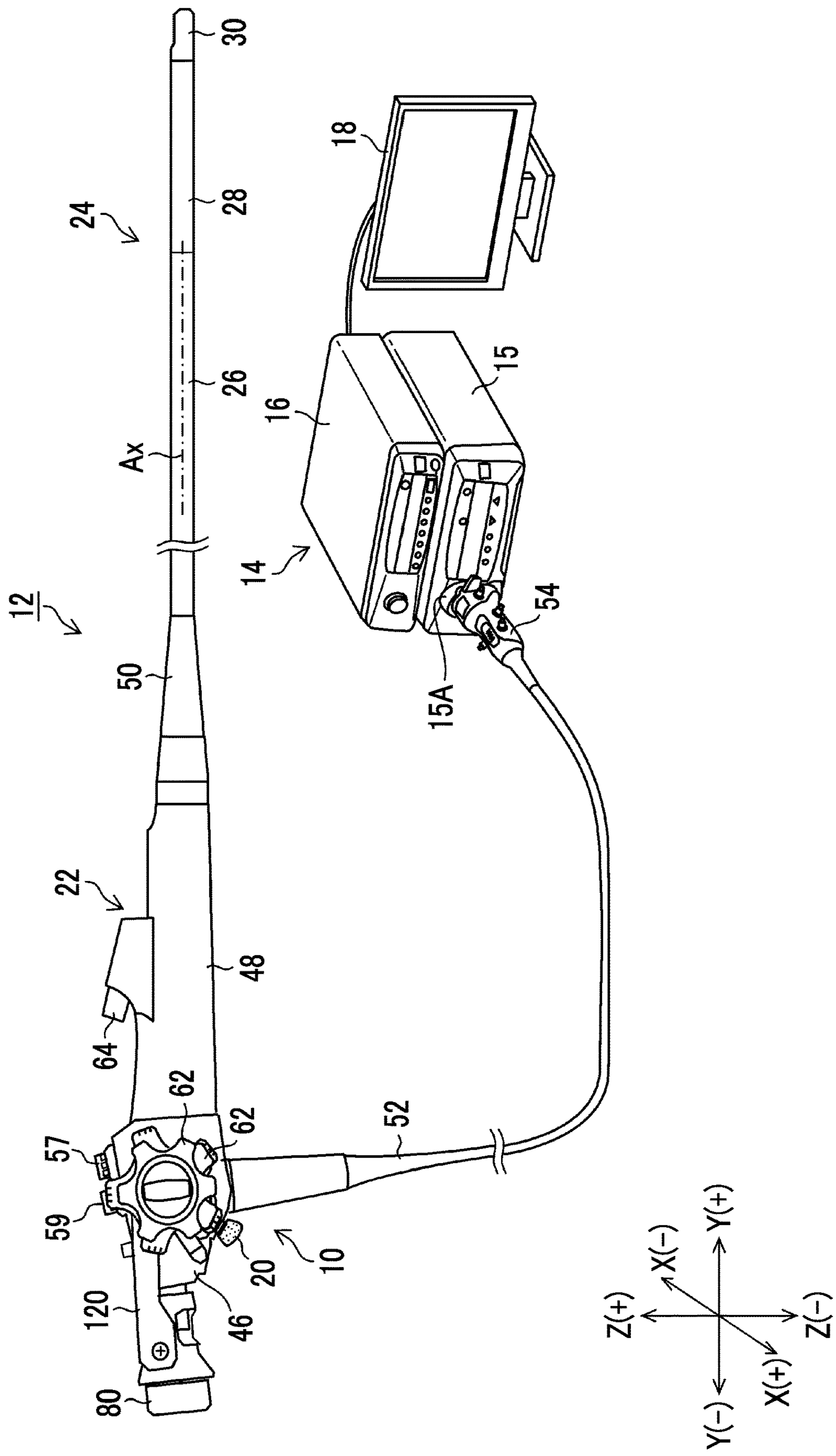


FIG. 1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Ax
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16
15
14
54
15A
12
50
22
48
64
62
62
57
59
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20
10
120
46
80
Y(+)
X(−)
Z(+)
Z(−)
X(+)
Y(−)

FIG. 2

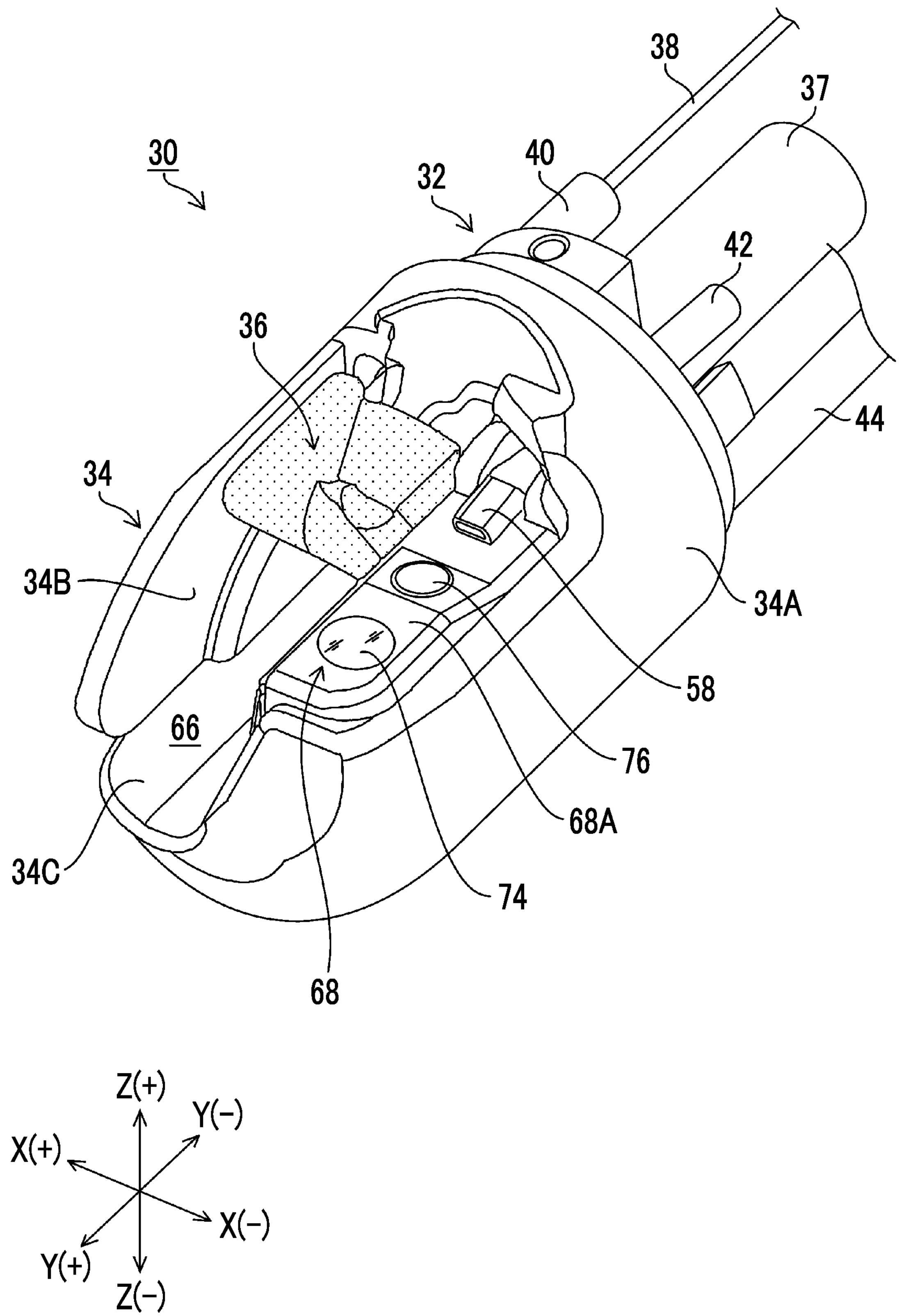
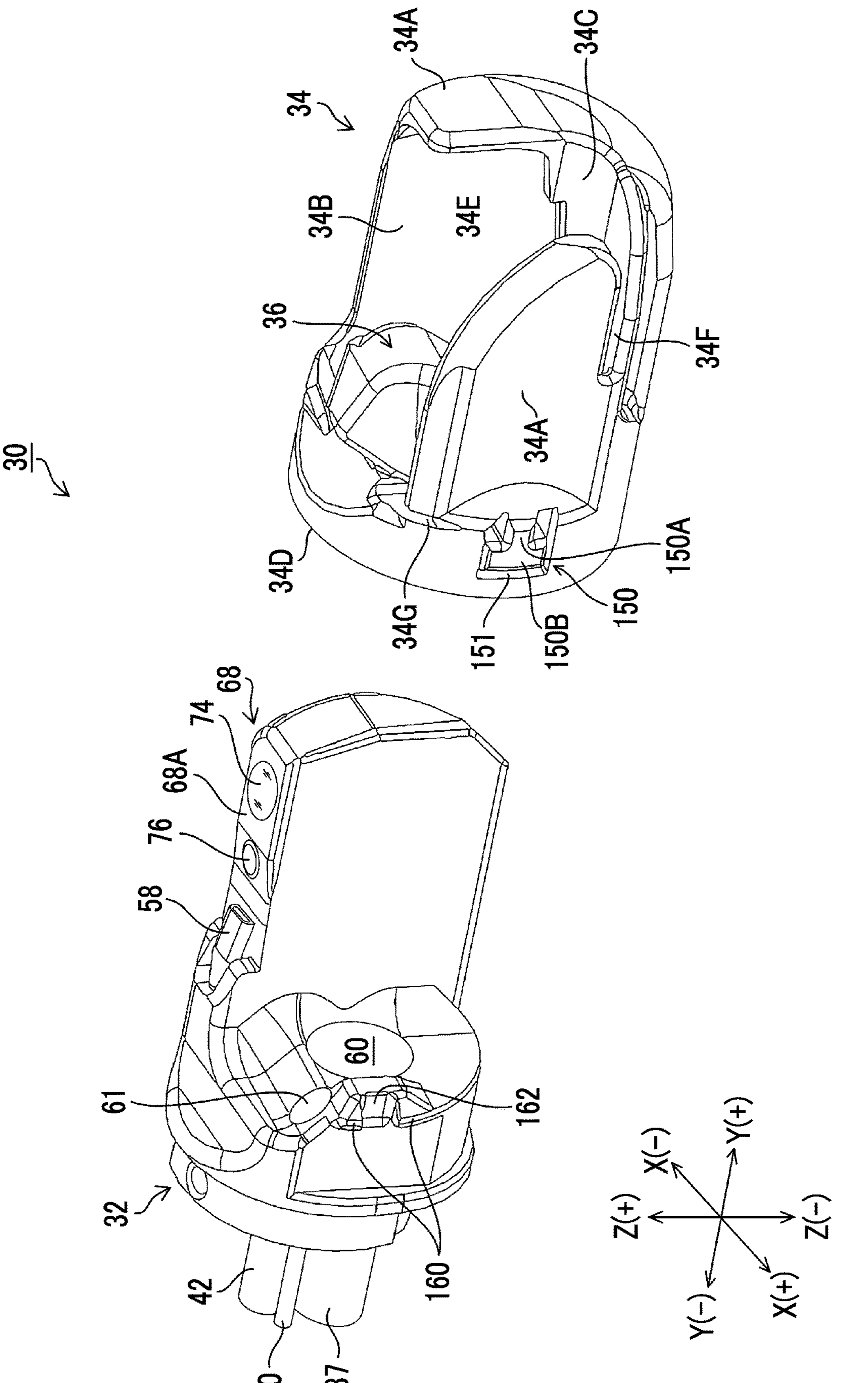


FIG. 3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34
34A
34B
34C
34E
36
34F
34A
34D
34G
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150B
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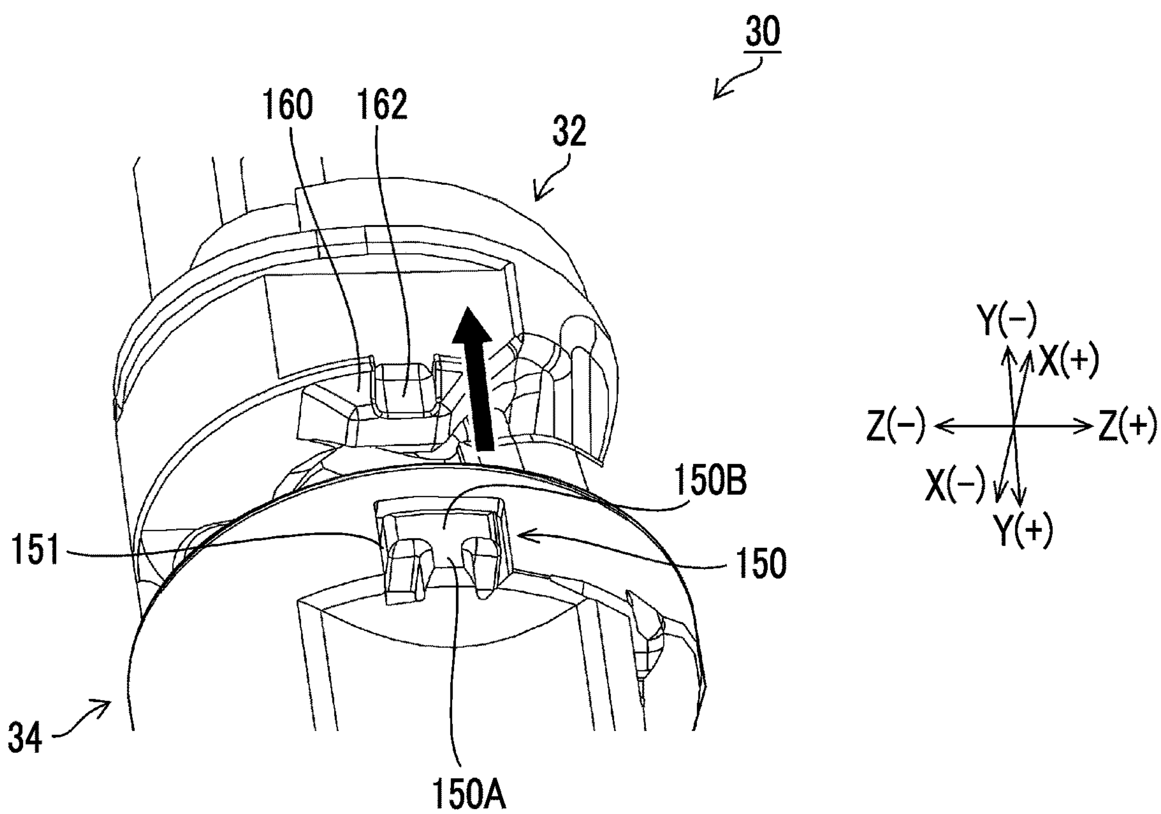
150A
68
74
68A
76
58
60
61
162
32
42
40
37
160
X(−)
Y(+)
Z(+)
Z(−)
Y(−)
X(+)

FIG. 4A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160
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32
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X(+)
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Z(+)
X(−)
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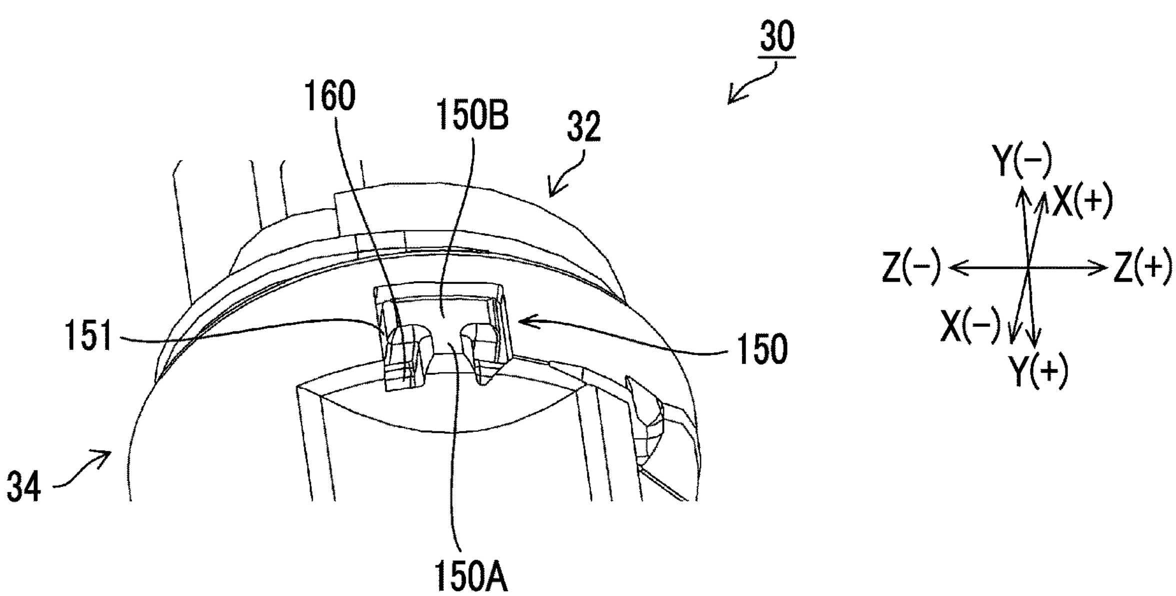
150A

FIG. 4B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Z(+)
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X(−)
Y(+)
34
150A

FIG. 5

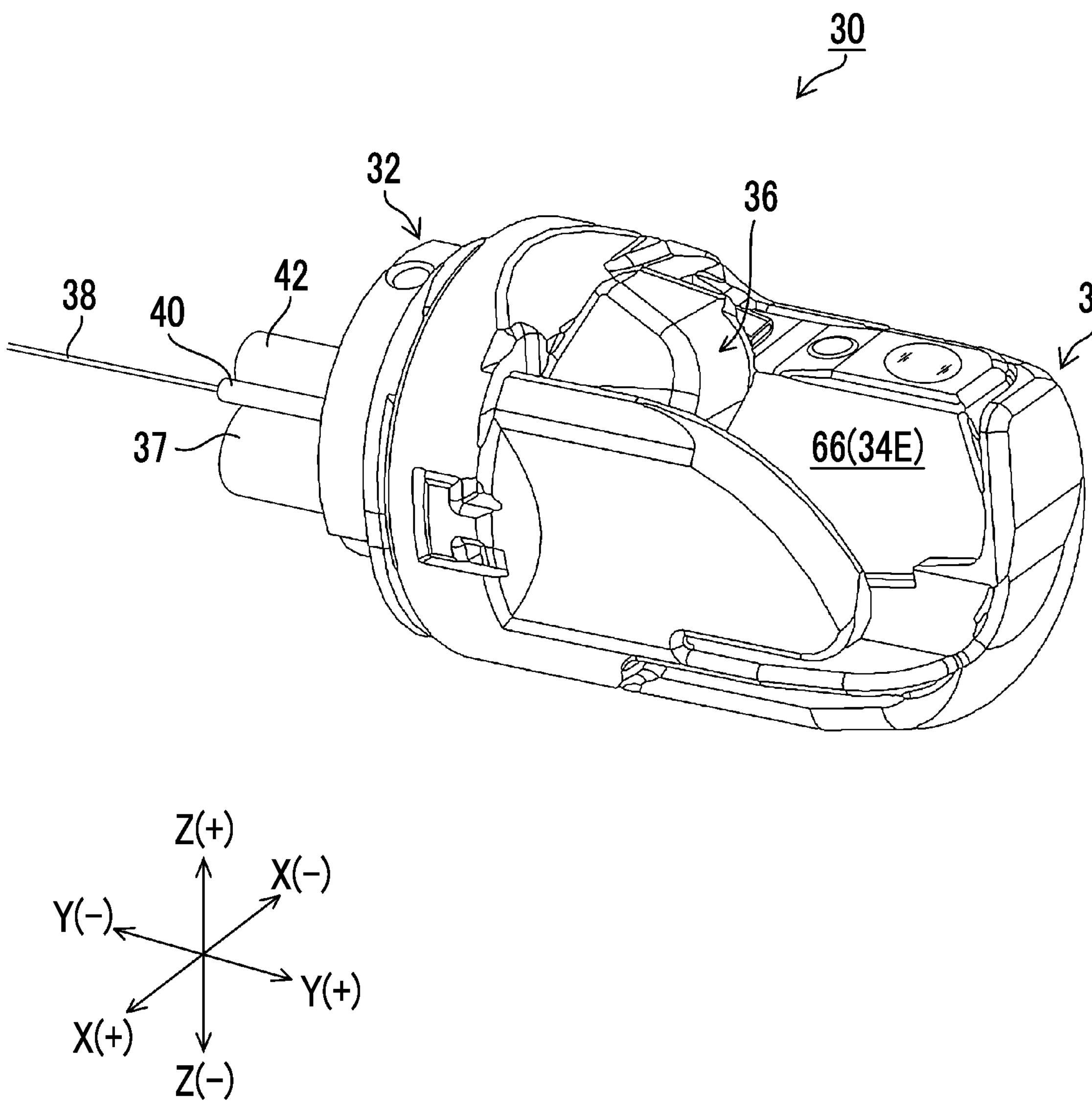


FIG. 6A

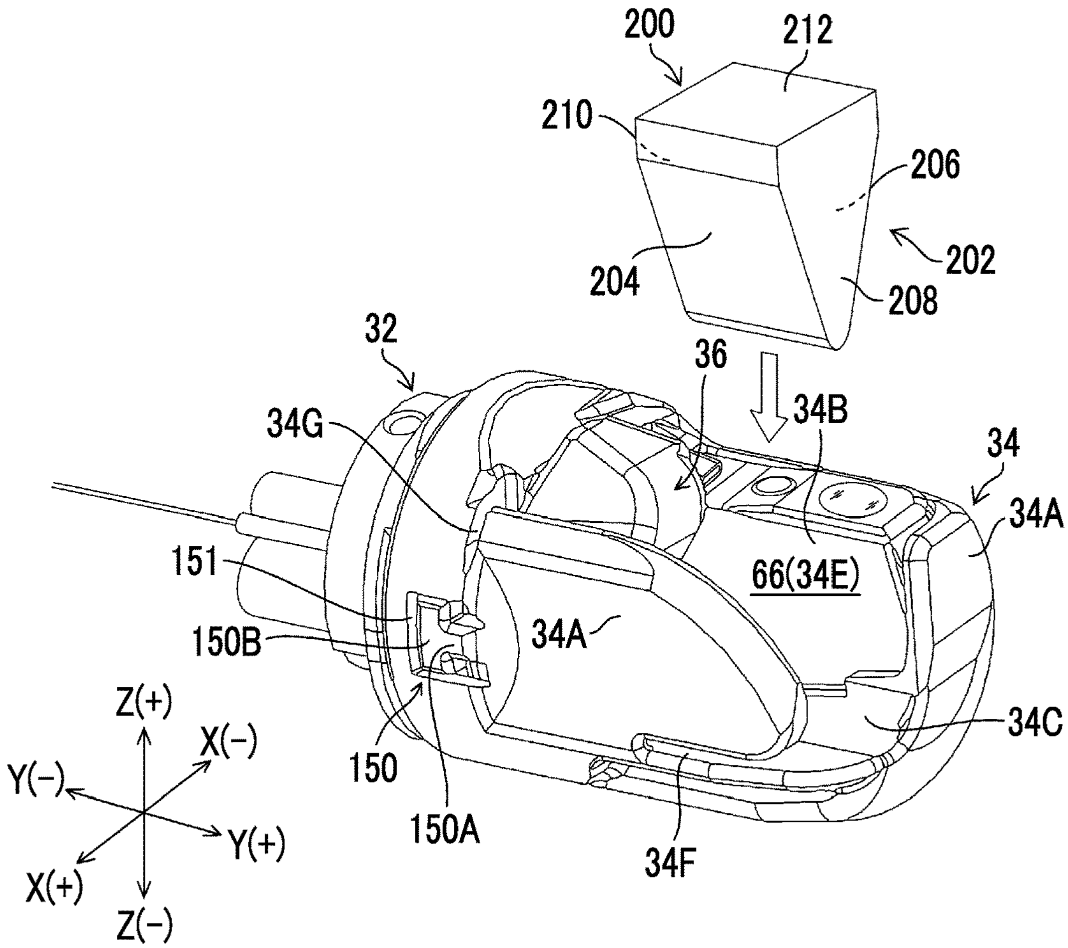
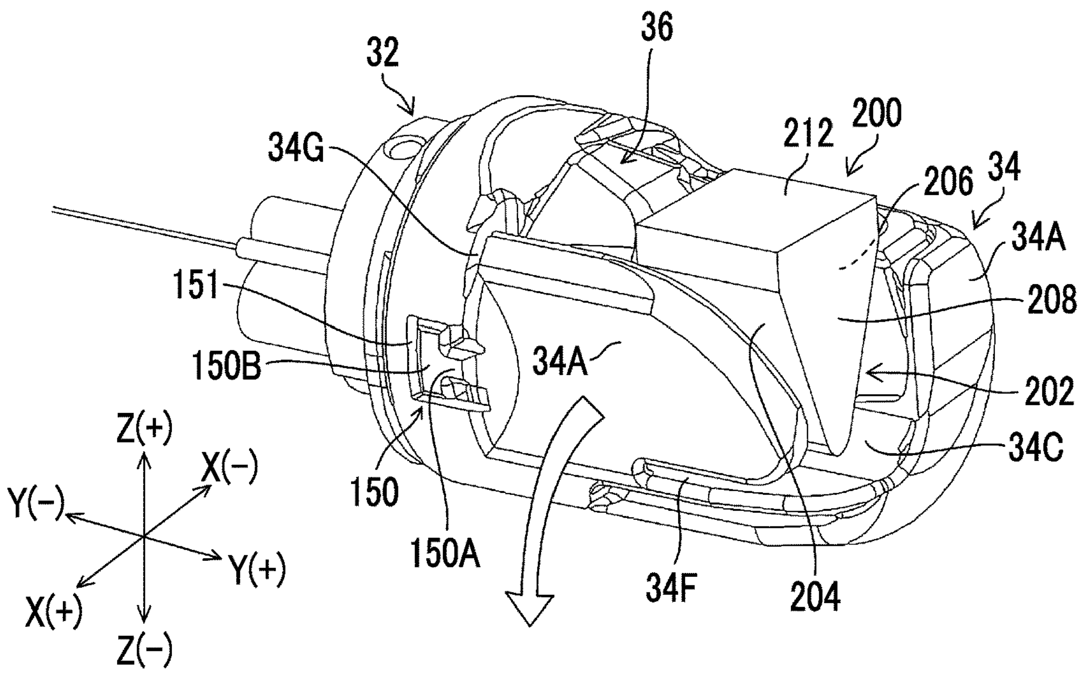

200
212
210
206
202
204
208
32
36
34B
34G
34
34A
66(34E)
151
34A
150B
34C
Z(+)
X(−)
Y(−)
150
Y(+)
150A
X(+)
34F
Z(−)

FIG. 6B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34
34A
208
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34A
150B
202
34C
Z(+)
X(−)
Y(−)
150
Y(+)
150A
X(+)
34F
204
Z(−)

FIG. 7

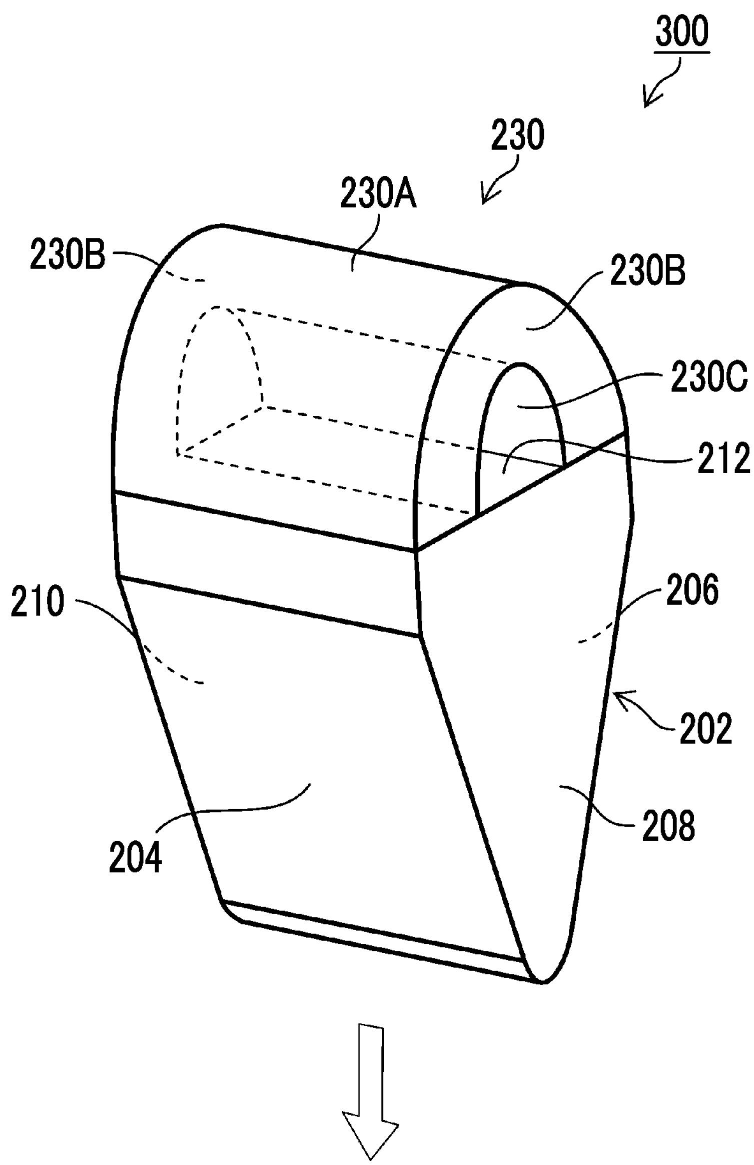
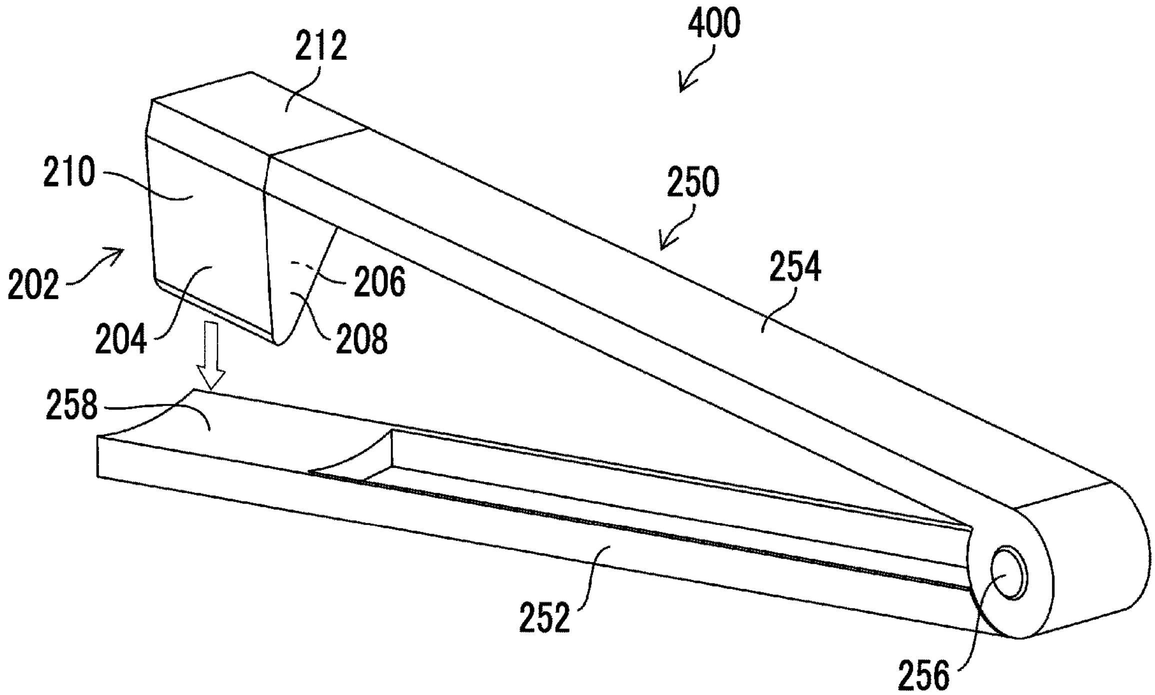


FIG. 8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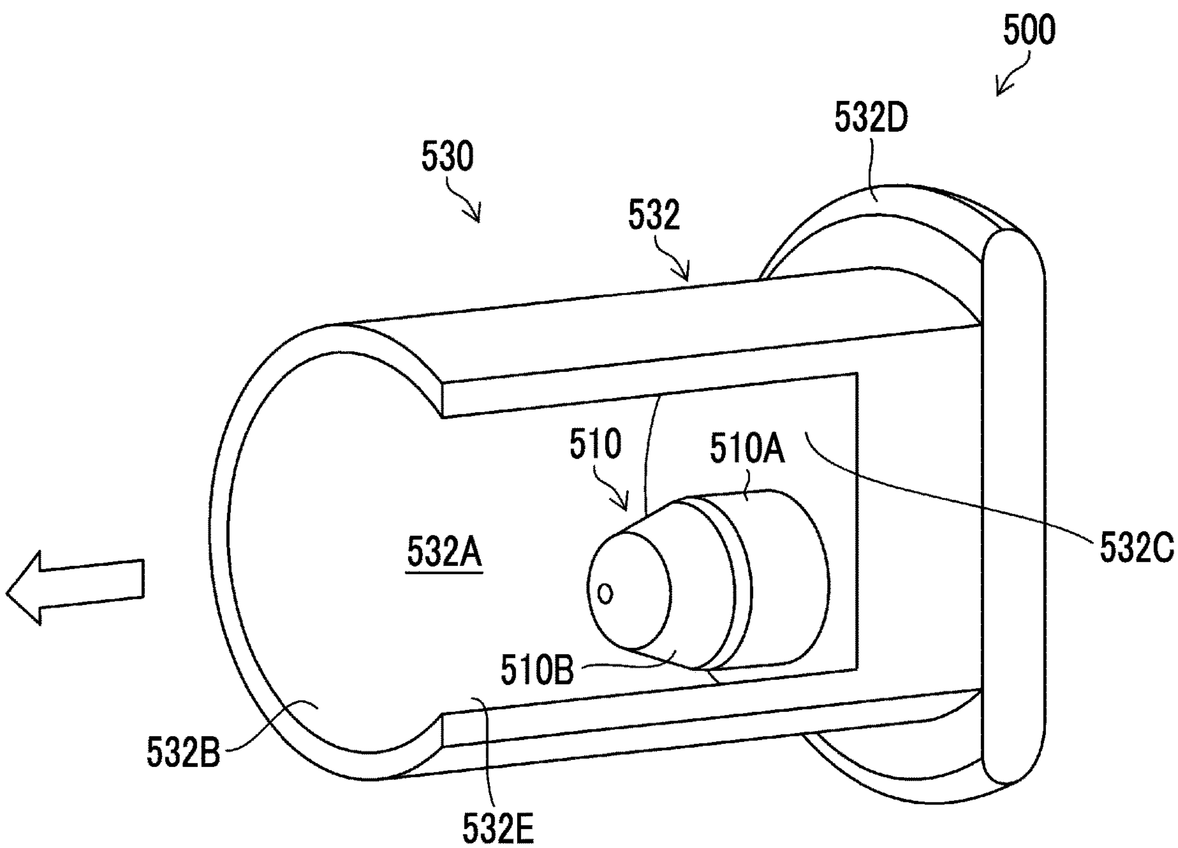
252
256

FIG. 9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532D
530
532
510
510A
532A
532C
510B
532B
532E

FIG. 10A

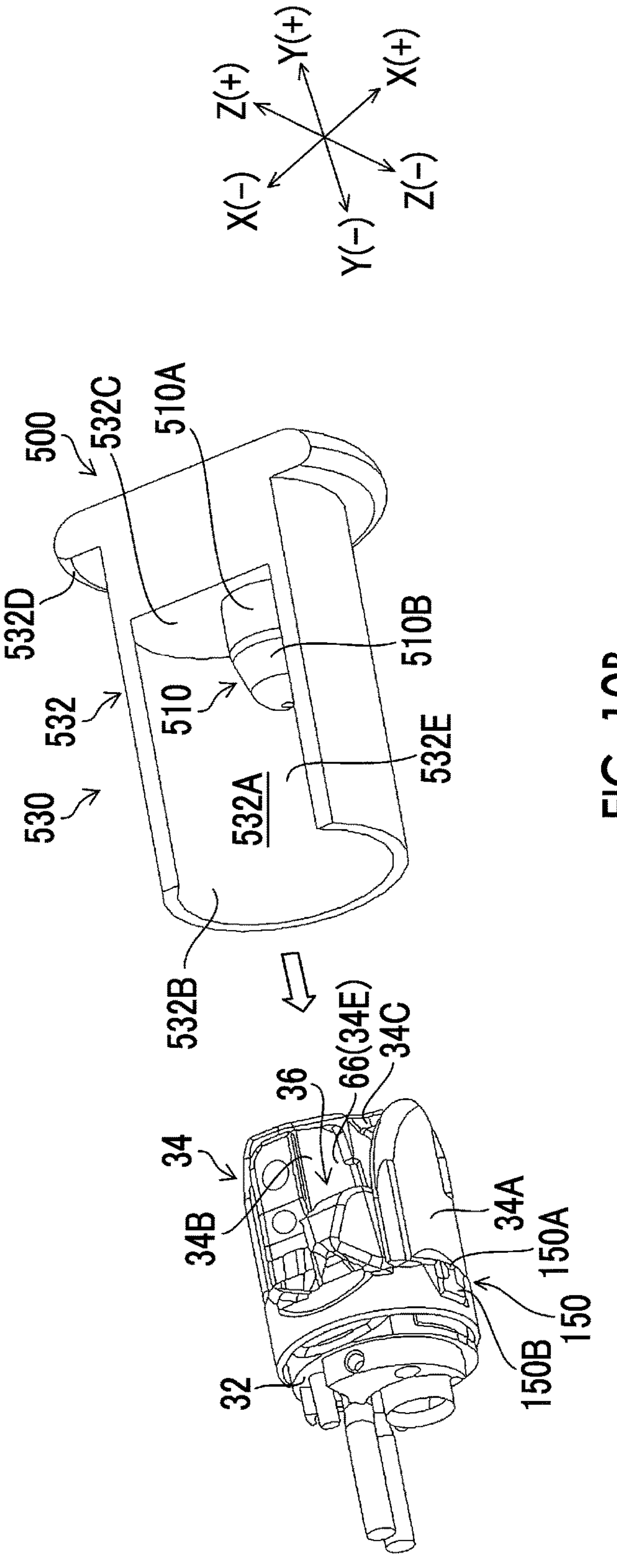


FIG. 10B

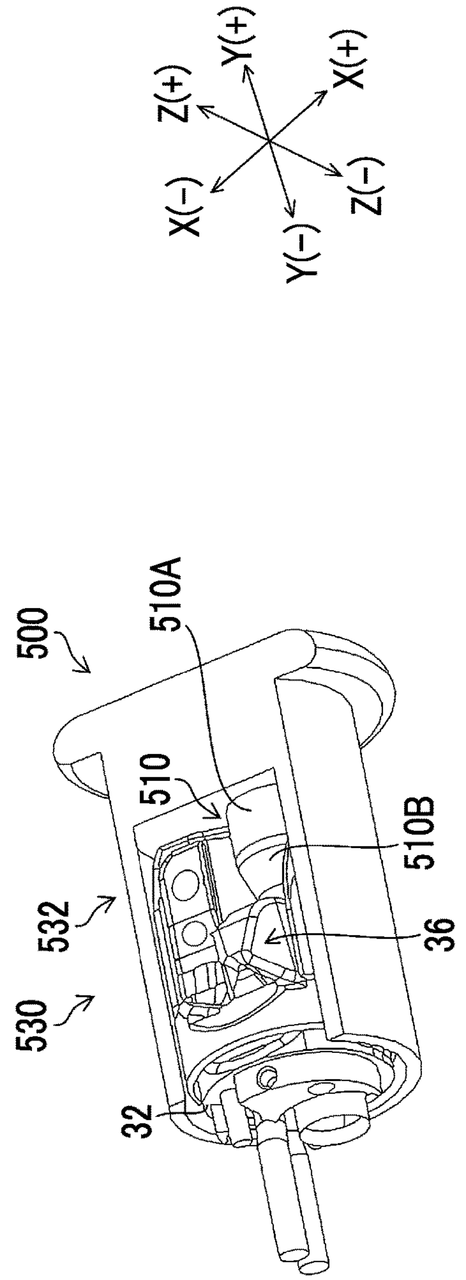
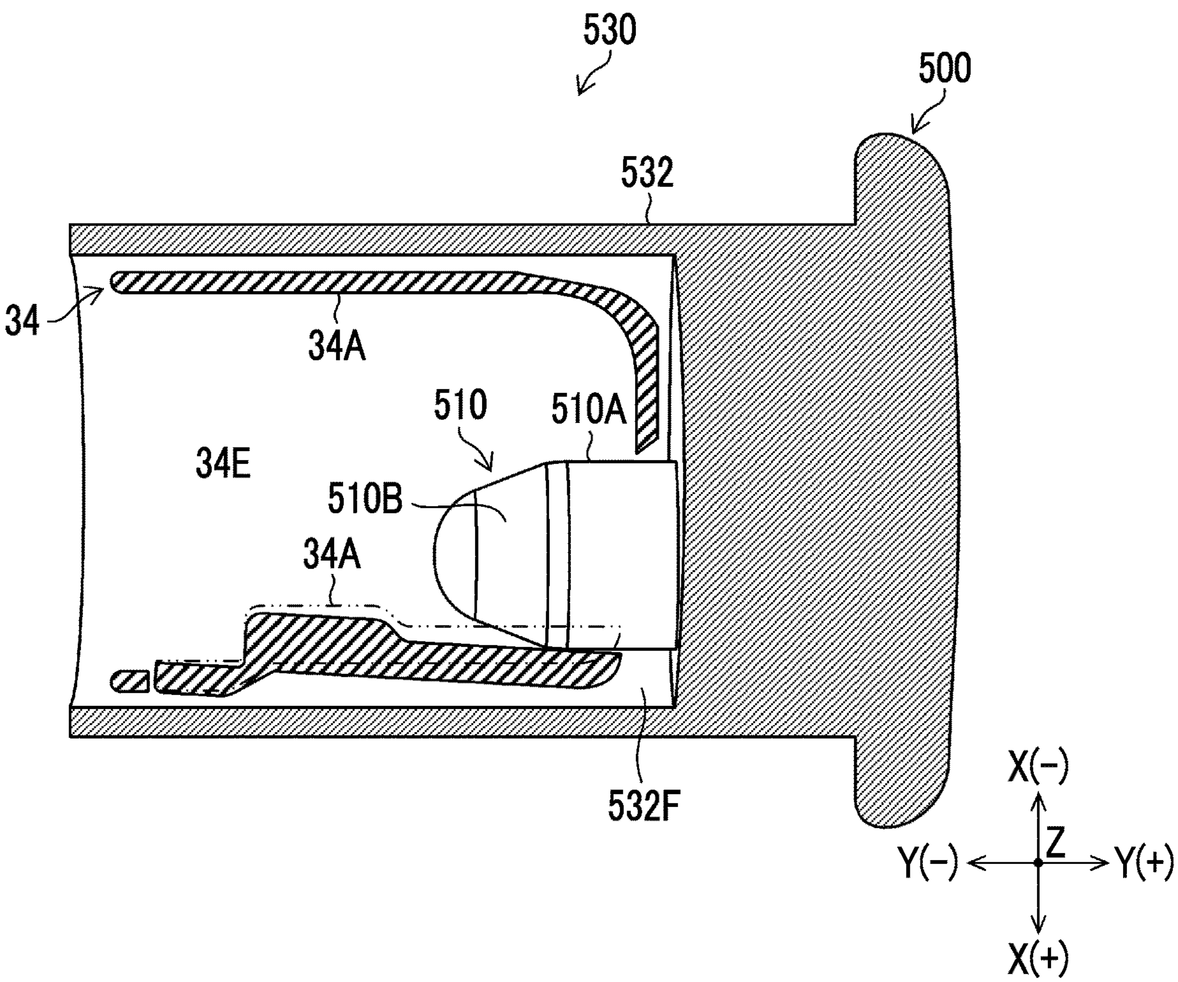


FIG. 11



DISTAL-END-CAP DETACHMENT JIG

CROSS-REFERENCE TO RELATED APPLICATION

The present application claims priority under 35 U.S.C § 119(a) to Japanese Patent Application No. 2020-142536 filed on Aug. 26, 2020, which is hereby expressly incorporated by reference, in its entirety, into the present application.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to a distal-end-cap detachment jig.

2. Description of the Related Art

In the endoscope, various treatment tools are introduced from the treatment tool inlet port provided in the operation part, and the treatment tools are led out of the treatment tool outlet port opened to the distal end of the insertion part to be used for treatment. For example, a treatment tool such as a guide wire or a contrast tube is used in a duodenal endoscope. A treatment tool such as a puncture needle is used in ultrasonic endoscope. A treatment tool such as forceps or a snare is used in other forward-viewing endoscopes and oblique-viewing endoscopes. In order to perform treatment at a desired position in an object to be examined, the lead-out direction of such a treatment tool needs to be changed at a distal end thereof. For this purpose, a distal-end-portion body of the distal end portion is provided with an elevator that changes the lead-out direction of the treatment tool. The endoscope is provided with a treatment-tool elevating mechanism that changes the posture of the elevator between an elevating position and a lying position.

The endoscope needs to be washed after the treatment. JP1995-184838 (JP-H07-184838), JP1997-075295 (JP-H09-075295), WO2017/122692A, and WO2018/051626A disclose that a distal end cap is attachably and detachably mounted on the distal-end-portion body provided with the elevator, and the distal end cap is detached through a jig after the treatment, in order to improve washability.

SUMMARY OF THE INVENTION

Incidentally, there is a concern that the endoscope may be damaged or a load may be applied to the endoscope when the distal end cap mounted on the distal-end-portion body is detached by the jig.

The present invention has been made in view of such circumstances, and an object thereof is to provide a distal-end-cap detachment jig capable of safely detaching the distal end cap from the distal-end-portion body.

A distal-end-cap detachment jig according to a first aspect that is used to detach a distal end cap mounted on a distal end of a side-viewing scope and having an inner space which communicates with a cap opening allowing a passage of a treatment tool, the distal-end-cap detachment jig comprises at least a body portion that has an inclined surface, in which the inclined surface is tapered in a direction of insertion into the inner space of the distal end cap, and in a case where the body portion is inserted into the inner space of the distal end cap, the inclined surface deforms the distal end cap in a direction of expanding the inner space of the distal end cap.

The distal-end-cap detachment jig according to a second aspect, further comprises a connection portion that is connected to the body portion, in which the connection portion is connected at a position opposite to the direction of insertion of the body portion and is provided with a finger hole extending in a direction orthogonal to the direction of insertion.

In the distal-end-cap detachment jig according to a third aspect, the finger hole is a through-hole that penetrates the connection portion.

The distal-end-cap detachment jig according to a fourth aspect, further comprises a connection portion that is connected to the body portion, in which the connection portion includes a fixed arm and a sliding arm that is connected to one end of the fixed arm through a fulcrum portion, the body portion is connected to an end part of the sliding arm on a side opposite to the fulcrum portion, and the body portion can be inserted into the inner space of the distal end cap with the fulcrum portion as a fulcrum.

In the distal-end-cap detachment jig according to a fifth aspect, a side of the fixed arm opposite to the fulcrum portion is formed by a curved surface that follows a shape of the distal end cap.

The distal-end-cap detachment jig according to a sixth aspect, further comprises a connection portion that is connected to the body portion, in which the connection portion has a housing member in which a space that houses the distal end cap is formed, the housing member has a first opening that allows housing the distal end cap and a bottom that faces the first opening, and the body portion is connected to the bottom at a position opposite to the direction of insertion of the body portion.

In the distal-end-cap detachment jig according to a seventh aspect, the space includes an escape space that houses the distal end cap deformed by the body portion.

In the distal-end-cap detachment jig according to an eighth aspect, the housing member has a gutter shape having a second opening that faces a side orthogonal to the direction of insertion of the body portion.

With the distal-end-cap detachment jig according to the aspects of the present invention, it is possible to suppress the damage or load applied to the endoscope.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a diagram showing a configuration of an endoscope system including an endoscope.

FIG. **2** is a perspective view of a distal end portion.

FIG. **3** is an exploded perspective view of the distal end portion.

FIGS. **4**A and **4**B are enlarged perspective views of the distal end portion.

FIG. **5** is a perspective view of the distal end portion when viewed from an angle different from that of FIG. **2**.

FIGS. **6**A and **6**B are perspective views showing a procedure for detaching the distal end cap using a first embodiment of a distal-end-cap detachment jig.

FIG. **7** is a perspective view showing a first modification example of the first embodiment of the distal-end-cap detachment jig.

FIG. **8** is a perspective view showing a second modification example of the first embodiment of the distal-end-cap detachment jig.

FIG. **9** is a perspective view of a second embodiment of the distal-end-cap detachment jig.

FIGS. **10**A and **10**B are perspective views showing a procedure for detaching the distal end cap using the second embodiment of the distal-end-cap detachment jig.

FIG. **11** is a cross-sectional view of the distal-end-cap detachment jig in a state in which the distal-end-cap detachment jig is inserted into the distal end cap.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

FIG. **1** is a diagram showing a configuration of an endoscope system **12** including an endoscope **10**. The endoscope system **12** comprises the endoscope **10**, an endoscope processor apparatus **14**, and a display **18**.

The endoscope **10** comprises a proximal operation part **22** (also referred to as an operation part) provided with an elevating operation lever **20**, and an insertion part **24** of which the proximal end is connected to the proximal operation part **22** and which is inserted into an object to be examined. The elevating operation lever **20** is an example of an operation member.

The insertion part **24** has a major axis direction Ax from the proximal end to the distal end, and comprises a soft portion **26**, a bendable portion **28**, and a distal end portion **30** in this order from the proximal end side to the distal end side. The schematic configuration of the distal end portion **30** will be described.

FIG. **2** is an enlarged perspective view of the distal end portion **30**. Here, the endoscope **10** (see FIG. **1**) is a side-viewing endoscope (also referred to as a side-viewing scope) that is used as, for example, a duodenal endoscope, and the distal end portion **30** of FIG. **2** has a configuration of a side-viewing endoscope.

In FIG. **2**, a distal end cap **34** is mounted on the distal-end-portion body **32**, whereby the distal end portion **30** is formed. A treatment tool elevator **36** (hereinafter, elevator **36**) is attached to the distal end cap **34**.

FIG. **2** shows various components provided in the insertion part **24** of the endoscope **10** (see FIG. **1**), in addition to the distal end portion **30**. A treatment tool channel **37** that guides the distal end of a treatment tool (not shown) to the distal-end-portion body **32**, an elevating operation wire **38** (hereinafter, referred to as a wire **38**) that is used to perform operation for changing the lead-out direction of the distal end of the treatment tool led out of the distal-end-portion body **32**, a wire channel **40** through which the wire **38** is inserted, an air/water supply tube **42**, and a cable insertion channel **44** are shown. An air/water supply nozzle **58** is connected to the air/water supply tube **42** and communicates with the air/water supply tube **42**. Further, components such as a light guide (not shown) that guides illumination light supplied from a light source device **15** (see FIG. **1**) to the distal-end-portion body **32** and an angle wire (not shown) that is used to perform operation for bending the bendable portion **28** (see FIG. **1**) are provided in the insertion part **24**. The wire **38** is inserted through the wire channel **40** and disposed therein so as to be able to advance and retreat. The wire channel **40** is disposed from the proximal operation part **22** (see FIG. **1**) to the insertion part **24**. An illumination window **74** that is used to emit light and an observation window **76** that is used to capture an observation image are provided on the front end face of the distal end portion **30**.

In the present specification, a three-dimensional Cartesian coordinate system including three-axis directions (an X-axis direction, a Y-axis direction, and a Z-axis direction) will be used to make a description. That is, in a case where the lead-out direction of the treatment tool (not shown) by the

elevator **36** when the distal end portion **30** is viewed from the proximal operation part **22** is set as an upward direction, the upward direction indicates a Z(+) direction and a downward direction which is a direction opposite to the upward direction indicates a Z(−) direction. Further, in this case, a right direction indicates an X(+) direction and a left direction indicates an X(−) direction. Furthermore, in this case, a forward direction (a direction toward the distal end in the major axis direction Ax of the insertion part **24**) indicates a Y(+) direction and a rearward direction (a direction toward the proximal end in the major axis direction Ax of the insertion part **24**) indicates a Y(−) direction. The Y-axis direction including the Y(+) direction and the Y(−) direction is a direction parallel to the major axis direction Ax of the insertion part **24**. The Z-axis direction is a direction orthogonal to the major axis direction Ax. The X-axis direction is a direction orthogonal to each of the Y-axis direction and the Z-axis direction.

As shown in FIG. **1**, the proximal operation part **22** is formed in a substantially cylindrical shape as a whole. The proximal operation part **22** has an operation-part body **46** provided with the elevating operation lever **20** and a grip portion **48** linked to the operation-part body **46**. The grip portion **48** is a portion that is gripped by the operator when the endoscope **10** is operated, and the proximal end of the insertion part **24** is connected to the distal end side of the grip portion **48** through a bending-proof pipe **50**. A link member **120** is disposed outside the operation-part body **46**. The link member **120** moves in conjunction with the operation of the elevating operation lever **20**. A wire fixing member **80** is housed and fixed on the proximal end side of the link member **120**. The wire fixing member **80** fixes the wire **38** (not shown). The wire fixing member **80** fixes the wire **38** and the link member **120**. That is, the elevating operation lever **20** and the wire **38** are mechanically connected to each other through the wire fixing member **80**. The wire fixing member **80** may comprise, for example, a collet. The collet has a plurality of gaps. The collet closes the gaps to grip and fix the wire **38**. The collet opens the gaps to release the fixing of the wire **38**. The opening and closing of the gaps of the collet may be linked to, for example, the rotational operation of the wire fixing member **80**.

The proximal end of a universal cable **52** is connected to the operation-part body **46**, and a connector device **54** is provided in the distal end of the universal cable **52**. The connector device **54** is connected to an endoscope processor apparatus **14**. The endoscope processor apparatus **14** comprises a light source device **15** and an image processing device **16**. The light source device **15** is provided with a processor-side connector **15**A to which the connector device **54** is connected. Further, the display **18** that displays an image which is subjected to image processing by the image processing device **16** is connected to the image processing device **16**. The endoscope system **12** has a configuration in which power, optical signals, and the like are transmitted between the endoscope **10** and the endoscope processor apparatus **14** in a non-contact manner through a connector portion that is constituted of the connector device **54** and the processor-side connector **15**A. Accordingly, the light from the light source device **15** is transmitted through an optical fiber cable (not shown) and is emitted from the illumination window **74** (see FIG. **2**) provided on the front end face of the distal end portion **30**. Further, the optical signal of the image captured from the observation window **76** (see FIG. **2**) is subjected to image processing by the image processing device **16** and is displayed as an image on the display **18**.

Further, an air/water supply button **57** and a suction button **59** are arranged side by side on the operation-part body **46**. The air/water supply button **57** is a button that can be operated in two stages, and air may be supplied to the air/water supply nozzle **58** (see FIG. **2**) through the air/water supply tube **42** by the first-stage operation, and water may be supplied to the air/water supply nozzle **58** through the air/water supply tube **42** by the second-stage operation. Further, in a case where the suction button **59** is operated, body fluid, such as blood, may be sucked from the treatment tool outlet port **60** (see FIG. **3**) through the treatment tool channel **37**.

As shown in FIG. **1**, a pair of angle knobs **62** and **62** that are used to perform operation for bending the bendable portion **28** are arranged on the operation-part body **46**. The pair of angle knobs **62** and **62** are coaxially provided so as to be rotationally movable. For example, four angle wires (not shown) are connected to the angle knobs **62** and **62**, and the bendable portion **28**, and the angle wires are pushed and pulled by the rotationally moving operation of the angle knobs **62** and **62**, so that the bendable portion **28** is vertically and laterally bent.

Further, the elevating operation lever **20** is rotatably provided coaxially with the angle knobs **62** and **62**. The elevating operation lever **20** is rotationally operated by the hand of the operator who grips the grip portion **48**. In a case where the elevating operation lever **20** is rotationally operated, the link member **120** moves, and the wire fixing member **80** fixed to the link member **120** moves. Since the wire **38** shown in FIG. **2** is fixed to the wire fixing member **80**, the wire **38** is pushed and pulled by this operation. The wire **38** is operated to be pushed and pulled, whereby the posture of the elevator **36** connected to the distal end of the wire **38** is changed between the lying position and the elevating position.

As shown in FIG. **1**, the grip portion **48** of the proximal operation part **22** comprises a treatment tool inlet port **64** into which the treatment tool is introduced. The treatment tool (not shown) of which the distal end as a leading end is introduced from the treatment tool inlet port **64** is inserted through the treatment tool channel **37** (see FIG. **3**) and is led out of the treatment tool outlet port **60** (see FIG. **3**). Examples of the treatment tool may include treatment tools such as biopsy forceps of which the distal end has a cup capable of collecting body tissue, a knife for endoscopic sphincterotomy (EST), or a contrast tube.

Next, the structure of the distal end portion **30** will be described with reference to FIGS. **2** and **3**. FIG. **2** is a perspective view of the distal end portion, and FIG. **3** is an exploded perspective view of the distal end portion.

The distal-end-portion body **32** is made of, for example, a metal material having corrosion resistance, and has a partition wall **68** provided in a protruding manner in the Y(+) direction as shown in FIGS. **2** and **3**.

The illumination window **74** and an observation window **76** are arranged adjacent to each other in the Y-axis direction on the upper surface **68**A on the Z(+) side of the partition wall **68**. The illumination window **74** can irradiate the visual field region in the Z(+) direction with illumination light, and the observation window **76** can observe the visual field region in the Z(+) direction. The distal-end-portion body **32** is provided with the air/water supply nozzle **58** facing the observation window **76**, and the observation window **76** is washed by air and water ejected from the air/water supply nozzle **58**.

The distal end cap **34** shown in FIG. **2** or **3** is made of an elastic material, for example, a rubber material, such as fluororubber or silicone rubber, and a resin material, such as polysulfone or polycarbonate.

The distal end cap **34** comprises a cap opening **34**B and a wall **34**A that defines a distal end opening **34**C continuous with the cap opening **34**B. The wall **34**A has a mounting opening **34**D, and the mounting opening **34**D allows the distal-end-portion body **32** to be inserted.

The distal end cap **34** is formed in a substantially tubular shape by the wall **34**A that defines the cap opening **34**B, the distal end opening **34**C, and the mounting opening **34**D, and an inner space **34**E is formed in the distal end cap **34**.

A through-hole **61** is formed in the distal-end-portion body **32**, and the wire **38** (not shown) is inserted through the through-hole **61**. The distal end of the wire **38** is connected to the elevator **36**.

As shown in FIG. **2**, when the distal end cap **34** is mounted on the distal-end-portion body **32**, a part of the inner space **34**E is occupied by the partition wall **68**. The elevator housing space **66** is defined by the partition wall **68** and the wall **34**A of the distal end cap **34**. The elevator housing space **66** is disposed at a position on the X(+) direction side of the partition wall **68** and a position on the Y(+) direction side of the treatment tool outlet port **60**.

As shown in FIG. **2**, when the distal end cap **34** is mounted on the distal-end-portion body **32**, the cap opening **34**B is directed in the Z(+) direction. The treatment tool outlet port **60** (see FIG. **3**) of the distal-end-portion body **32** communicates with the cap opening **34**B through the elevator housing space **66**.

The elevator **36** is rotatably supported in the inner space **34**E of the distal end cap **34**. A rotating shaft (not shown) is attached to the elevator **36**. The rotating shaft is attached to the elevator **36** on the side opposite to the position facing the cap opening **34**B. A bearing (not shown) that rotatably supports the rotating shaft of the elevator **36** is disposed in the inner space **34**E of the distal end cap **34**.

When the wire **38** shown in FIG. **2** is pushed and pulled, the elevator **36** is rotated about the rotating shaft, and the posture thereof is changed between the lying position and the elevating position.

The distal end cap **34** has a configuration in which the elevator **36** to which the wire **38** is connected is attached in advance. When the treatment using the endoscope **10** ends, the distal end cap **34** formed in this way is detached from the distal-end-portion body **32** as will be described later, and is discarded together with the elevator **36** and the wire **38** as, for example, a disposable member. The elevator **36** may be attached to the distal-end-portion body **32** instead of the distal end cap **34**.

Next, mounting the distal end cap **34** on the distal-end-portion body **32** will be described with reference to FIGS. **3** to **5**.

As shown in FIG. **3**, the wall **34**A of the distal end cap **34** faces the X(+) direction and the X(−) direction with the cap opening **34**B interposed therebetween. The cantilever piece **150** is formed on the wall **34**A opposite to the side where the partition wall **68** is inserted. A cutout **151** is provided in the distal end cap **34**, whereby the cantilever piece **150** is formed. The cutout **151** penetrates the outside and the inside of the distal end cap **34**.

The cantilever piece **150** comprises a support piece **150**A and a stopped portion **150**B connected to the support piece **150**A. The side of the support piece **150**A opposite to the stopped portion **150**B is connected to the wall **34**A to form a fixed end of the cantilever piece **150**. The stopped portion **150**B of the cantilever piece **150** is a free end that is not connected to the wall **34**A. The cantilever piece **150** extends along the Y-axis direction, and the fixed end is positioned on the distal end side (Y(+) direction) than the free end.

The stopped portion **150**B is larger in width in the Z-axis direction than the support piece **150**A. When viewed in the X(+) direction, the cantilever piece **150** has a T-shape as a whole.

Since the support piece **150**A is bent and deformed with the fixed end as a fulcrum, the stopped portion **150**B, which is a free end, can be displaced in the X(+) direction and the X(−) direction.

As shown in FIG. **3**, a first cut **34**F and a second cut **34**G are formed in the distal end cap **34**. The first cut **34**F is continuous with the distal end opening **34**C and extends in the direction along the Y-axis direction. The first cut **34**F penetrates the outside and the inside of the wall **34**A of the distal end cap **34**. The distal end cap **34** is divided into the Z(+) side and the Z(−) side with the first cut **34**F as a boundary. The first cut **34**F may not be completely parallel to the Y-axis direction as long as the first cut **34**F extends along the Y-axis direction.

The second cut **34**G extends in a direction orthogonal to the first cut **34**F. The second cut **34**G penetrates the outside and the inside of the wall **34**A of the distal end cap **34**. The second cut **34**G is continuous with the cutout **151**, extends in the Z(+) direction, and reaches the cap opening **34**B. The distal end cap **34** is divided into the Y(+) side and the Y(−) side with the second cut **34**G as a boundary. The first cut **34**F and the second cut **34**G have a positional relationship in which the first cut **34**F and the second cut **34**G are orthogonal to each other. Being orthogonal includes being completely orthogonal and being substantially orthogonal.

As shown in FIG. **3**, the distal-end-portion body **32** has two stopper portions **160** on the side facing the cantilever piece **150**. The two stopper portions **160** protrude in the X(+) direction. Further, the two stopper portions **160** are arranged along the Z-axis direction. A groove portion **162** is defined by the two stopper portions **160**. The two stopper portions **160** are arranged at an interval that is narrower than the width of the stopped portion **150**B and wider than the width of the support piece **150**A.

FIGS. **4**A and **4**B are enlarged perspective views of the distal end portion **30**, and FIG. **4**A shows a state immediately before the distal end cap **34** is mounted on the distal-end-portion body **32** and FIG. **4**B shows a state after the distal end cap **34** is mounted on the distal-end-portion body **32**. When the distal end cap **34** is mounted on the distal-end-portion body **32**, the distal end cap **34** moves toward the distal-end-portion body **32** as shown in FIG. **4**A.

As the distal end cap **34** moves, the stopped portion **150**B of the cantilever piece **150** comes into contact with the two stopper portions **160**. Further, when the distal end cap **34** moves toward the distal-end-portion body **32**, the support piece **150**A is bent and deformed in the X(+) direction with the fixed end as a fulcrum. The stopped portion **150**B, which is a free end, moves in a direction of running on the two stopper portions **160**.

As shown in FIG. **4**B, when the distal end cap **34** moves to the mounting position with respect to the distal-end-portion body **32**, the stopped portion **150**B goes over the two stopper portions **160** and moves to the proximal end side (Y(−)) than the two stopper portions **160**. The support piece **150**A returns to the original state due to elastic deformation. As shown in FIG. **4**A, since the interval between the two stopper portions **160** is narrower than the width of the stopped portion **150**B, the two stopper portions **160** and the stopped portion **150**B are engaged with each other. On the other hand, the support piece **150**A is housed in the groove portion **162**. The engagement between the stopper portions **160** and the stopped portion **150**B prevents the distal end cap **34** from moving from the distal-end-portion body **32** in the Y(+) direction and falling out.

As shown in FIG. **5**, the distal end cap **34** is mounted on the distal-end-portion body **32**. In this state, the distal end portion **30** of the endoscope **10** is inserted into the object to be examined. The treatment tool is directed in a desired lead-out direction in the object to be examined by the elevator **36** of the distal end portion **30**. After the treatment, the distal end cap **34** is detached and the distal end portion **30** is washed. Accordingly, it is required that the distal end cap **34** can be easily detached from the distal-end-portion body **32** and the damage or load applied to the endoscope **10** is suppressed.

Hereinafter, the distal-end-cap detachment jig will be described with reference to FIGS. **6**A to **11**.

First Embodiment

FIGS. **6**A and **6**B are perspective views showing a procedure for detaching the distal end cap using a first embodiment of the distal-end-cap detachment jig. FIG. **6**A is a perspective view showing a state before a distal-end-cap detachment jig **200** is inserted into the distal end cap **34**, and FIG. **6**B is a perspective view showing a state after the distal-end-cap detachment jig **200** is inserted into the distal end cap **34**.

As shown in FIG. **6**A, the distal-end-cap detachment jig **200** comprises a body portion **202**. The body portion **202** has two inclined surfaces **204** and **206** that face each other. The two inclined surfaces **204** and **206** are planes and are inclined with respect to the direction of insertion indicated by the arrow. The two inclined surfaces **204** and **206** intersect in the direction of insertion and form a tapered shape in the direction of insertion. The line of intersection between the inclined surfaces **204** and **206** extends in a straight line.

The body portion **202** comprises two planes **208** and **210** which face each other and connect the two inclined surfaces **204** and **206** to each other. The body portion **202** comprises a plane **212** parallel to the X-axis direction and the Y-axis direction, on the side opposite to the direction of insertion. It is preferable that the body portion **202** is integrally molded.

Since the body portion **202** of the distal-end-cap detachment jig **200** deforms the distal end cap **34**, the body portion **202** is preferably made of a material harder than the distal end cap **34**. For example, the body portion **202** is made of a resin material, a metal material, or the like.

As shown in FIG. **6**A, the distal-end-cap detachment jig **200** is prepared. The position of the distal-end-cap detachment jig **200** is adjusted with respect to the distal end cap **34** in a direction in which the tapered part of the body portion **202** faces the cap opening **34**B and at least one inclined surface **204** comes into contact with the wall **34**A. The line of intersection between the two inclined surfaces **204** and **206** is positioned in a direction substantially parallel to the Y-axis direction. When the distal end cap **34** is mounted on the distal-end-portion body **32**, the elevator housing space **66** and the inner space **34**E of the distal end cap **34** substantially coincide with each other.

As shown in FIG. **6**B, the body portion **202** is inserted into the inner space **34**E of the distal end cap **34** through the cap opening **34**B. For example, the plane **212** is pressed by a finger or the like, and the body portion **202** is pushed into the inner space **34**E of the distal end cap **34**.

The distance between the inclined surfaces **204** and **206** increases in a direction opposite to the direction of insertion. The distance between the inclined surfaces **204** and **206** at a certain position is larger than the distance between the partition wall **68** and the wall **34**A positioned on the X(+) side with respect to the partition wall **68**.

As the body portion **202** is inserted into the inner space **34**E of the distal end cap **34**, the inclined surface **204** pushes the wall **34**A in the direction (X(+) direction) of expanding the inner space **34**E of the distal end cap **34**, to deform the distal end cap **34**. The deformation of the distal end cap **34** facilitates the detachment of the distal end cap **34** from the distal-end-portion body **32**. A load applied to the bendable portion **28** (not shown) can be suppressed.

In FIG. **6**B, the deformation of the distal end cap **34** facilitates the release of the engaged stopped portion **150**B and stopper portions **160**.

In FIG. **6**B, since the first cut **34**F and the second cut **34**G orthogonal to the first cut **34**F are formed in the distal end cap **34**, the wall **34**A can be easily deformed.

The distal end cap **34** detached from the distal-end-portion body **32** is discarded without being reused.

Next, a first modification example of the first embodiment of a distal-end-cap detachment jig **300** will be described with reference to FIG. **7**. The distal-end-cap detachment jig **300** comprises a connection portion **230** connected to the body portion **202**. The connection portion **230** is connected to a plane **212** at a position opposite to the direction of insertion of the body portion **202**. The connection portion **230** has an outer form **230**A having a substantially semi-cylindrical shape and has a finger hole **230**C penetrating two bottom surfaces **230**B that face each other. The finger hole **230**C extends in a direction orthogonal to the direction of insertion indicated by the arrow, and in FIG. **7**, a direction along the line of intersection between the inclined surfaces **204** and **206**.

In the first modification example, the operator handles the distal-end-cap detachment jig **300** in a state in which a finger is inserted into the finger hole **230**C of the connection portion **230**. The operator can easily insert the body portion **202** into the inner space **34**E (see FIG. **6**) of the distal end cap **34**.

In the distal-end-cap detachment jig **300**, the connection of the body portion **202** and the connection portion **230** includes a case where the body portion **202** and the connection portion **230** are integrally molded and a case where the body portion **202** and the connection portion **230** are formed as separate members and are bonded to each other using an adhesive or the like.

The case where the finger hole **230**C penetrates is illustrated in FIG. **7**, but the finger hole **230**C does not need to penetrate as long as the finger can be inserted.

Next, a second modification example of the first embodiment of a distal-end-cap detachment jig **400** will be described with reference to FIG. **8**. The distal-end-cap detachment jig **400** comprises a connection portion **250** connected to the body portion **202**.

The connection portion **250** comprises a fixed arm **252**, a sliding arm **254**, and a fulcrum portion **256** that connects one end of the fixed arm **252** to one end of the sliding arm **254**.

The body portion **202** is connected to the end part of the sliding arm **254** on the side opposite to the fulcrum portion **256**. The direction of insertion of the body portion **202** is directed to the fixed arm **252** side.

A supporting surface **258** that is used to support the distal-end-portion body **32** on which the distal end cap **34** is mounted is provided on the side of the fixed arm **252** opposite to the fulcrum portion **256**. The supporting surface **258** is preferably a curved surface that follows the shape of the distal end cap **34**. The supporting surface **258** can stably support the distal-end-portion body **32** on which the distal end cap **34** is mounted.

The distal end cap **34** is detached by, for example, the following procedure. The distal-end-portion body **32** on which the distal end cap **34** is mounted is placed on the supporting surface **258** of the fixed arm **252** so that the distal end side faces the fulcrum portion **256**. The operator applies a force to the sliding arm **254** toward the fixed arm **252**. The body portion **202** is inserted into the inner space **34**E of the distal end cap **34** (not shown), and as in the first embodiment, the inclined surface **204** pushes the wall **34**A in the direction of expanding the inner space **34**E of the distal end cap **34**, to deform the distal end cap **34**.

The fulcrum portion **256** in FIG. **8** is formed of respective through-holes provided at one end of the fixed arm **252** and one end of the sliding arm **254** and a pin inserted into both the through-holes at the same time. In the fulcrum portion **256**, it is preferable that a biasing force is applied by a spring element (not shown) in a direction in which the fixed arm **252** and the sliding arm **254** are spaced apart from each other.

The fixed arm **252**, the sliding arm **254**, and the fulcrum portion **256** may be integrally molded. For example, a strip-shaped metal member is bent at substantially the center thereof, whereby the connection portion **250** may be formed. The bent portion forms the fulcrum portion.

Second Embodiment

FIG. **9** is a perspective view showing a second embodiment of the distal-end-cap detachment jig. A distal-end-cap detachment jig **500** comprises a body portion **510** and a connection portion **530** connected to the body portion **510**.

The body portion **510** has a shape different from the body portion **202** of the first embodiment. The body portion **510** comprises a cylindrical part **510**A and a distal end part **510**B that is continuous with the cylindrical part **510**A and that becomes tapered as the distal end part **510**B becomes farther from the cylindrical part **510**A. The distal end part **510**B has a substantially conical shape, and the side surface thereof forms an inclined surface that is tapered in the direction of insertion indicated by the arrow. The distal end shape of the distal end part **510**B may be a dome shape formed by a curved surface as shown in FIG. **9**, a flat surface, or a needle shape that converges to one point as long as the distal end part **510**B comprises the inclined surface that is tapered in the direction of insertion.

The connection portion **530** comprises a housing member **532** in which a space **532**A housing the distal end cap **34** is formed. The housing member **532** has a first opening **532**B that allows housing the distal end cap **34**, and a bottom **532**C that faces the first opening **532**B.

The body portion **510** is connected to the bottom **532**C of the housing member **532** at a position opposite to the direction of insertion of the body portion **510**. The housing member **532** comprises a flange **532**D on the side opposite to the first opening **532**B.

The housing member **532** shown in FIG. **9** comprises preferably a second opening **532**E facing the side orthogonal to the direction of insertion of the body portion **510**. Since the second opening **532**E is formed, the housing member **532** has a substantially C-shaped gutter shape in a cross-section taken along a direction orthogonal to the direction of insertion.

In the distal-end-cap detachment jig **500**, the connection of the body portion **510** and the connection portion **530** includes a case where the body portion **510** and the connection portion **530** are integrally molded and a case where the body portion **510** and the connection portion **530** are formed as separate members and are bonded to each other using an adhesive or the like.

The distal-end-cap detachment jig **500** may be made of the same material as in the first embodiment.

FIGS. **10**A and **10**B are perspective views showing a procedure for detaching the distal end cap using the second embodiment of the distal-end-cap detachment jig. FIG. **10**A is a perspective view showing a state before the distal-end-cap detachment jig **500** is inserted into the distal end cap **34**, and FIG. **10**B is a perspective view showing a state after the distal-end-cap detachment jig **500** is inserted into the distal end cap **34**.

As shown in FIG. **10**A, the distal-end-cap detachment jig **500** is prepared. The position of the distal-end-cap detachment jig **500** is adjusted with respect to the distal end cap **34** in a direction in which the tapered part of the body portion **510** faces the distal end opening **34**C and at least the inclined surface included in the distal end part **510**B comes into contact with the wall **34**A. The distal-end-cap detachment jig **500** is disposed in a state in which the first opening **532**B of the housing member **532** faces the distal end cap **34** side and the second opening **532**E of the housing member **532** faces the Z(+) side. As in the first embodiment, the elevator housing space **66** and the inner space **34**E of the distal end cap **34** substantially coincide with each other.

As shown in FIG. **10**B, the distal-end-portion body **32** on which the distal end cap **34** is mounted is housed in a space **532**A of the housing member **532**, and the body portion **510** is inserted into the inner space **34**E of the distal end cap **34** through the distal end opening **34**C. For example, the body portion **510** is pushed into the inner space **34**E of the distal end cap **34** while the housing member **532** is held by a finger or the like.

The diameter of the inclined surface included in the distal end part **510**B increases in the direction opposite to the direction of insertion. The outer diameter of the distal end part **510**B formed by the inclined surface at a certain position is larger than the distance between the partition wall **68** and the wall **34**A positioned on the X(+) side with respect to the partition wall **68**.

As the body portion **510** is inserted into the inner space **34**E of the distal end cap **34**, the inclined surface of the distal end part **510**B pushes the wall **34**A in the direction (X(+) direction) of expanding the inner space **34**E of the distal end cap **34**, to deform the distal end cap **34**. The deformation of the distal end cap **34** facilitates the detachment of the distal end cap **34** from the distal-end-portion body **32**. Damage to the observation window **76** of the distal-end-portion body **32** can be suppressed.

In FIG. **10**B, the operator can visually recognize the deformed state of the distal end cap **34** through the second opening **532**E of the housing member **532**.

Further, the deformation of the distal end cap **34** facilitates the release of the engaged stopped portion **150**B and stopper portions **160**.

Further, since the first cut **34**F and the second cut **34**G are formed in the distal end cap **34**, the wall **34**A can be easily deformed.

The distal end cap **34** detached from the distal-end-portion body **32** is discarded without being reused.

FIG. **11** is a cross-sectional view that is taken along a plane parallel to the X-axis direction and the Y-axis direction and passing through the substantially center of the body portion **510**, in the state of FIG. **10**B. As shown in FIG. **11**, when the body portion **510** is inserted into the inner space **34**E of the distal end cap **34**, the inclined surface of the distal end part **510**B pushes the wall **34**A in the X(+) direction to deform the distal end cap **34**.

In order to make the distal end cap **34** deformable, the space **532**A has a size in which the space **532**A includes an escape space **532**F between the wall **34**A of the distal end cap **34** and the interior wall of the housing member **532**.

In the first and second embodiments, the independent distal-end-cap detachment jig has been exemplified. However, the distal-end-cap detachment jig may be provided on a mounting part mounted on the operation part of the side-viewing scope. In this case, it is not necessary to wash and disinfect the distal-end-cap detachment jig itself. Further, the risk of losing the distal-end-cap detachment jig can be avoided.

EXPLANATION OF REFERENCES

10: endoscope
12: endoscope system
14: endoscope processor apparatus
15: light source device
15A: processor-side connector
16: image processing device
18: display
20: elevating operation lever
22: proximal operation part
24: insertion part
26: soft portion
28: bendable portion
30: distal end portion
32: distal-end-portion body
34: distal end cap
34A: wall
34B: cap opening
34C: distal end opening
34D: mounting opening
34E: inner space
34F: first cut
34G: second cut
36: treatment tool elevator
37: treatment tool channel
38: elevating operation wire
40: wire channel
42: air/water supply tube
44: cable insertion channel
46: operation-part body
48: grip portion
50: bending-proof pipe
52: universal cable
54: connector device
57: air/water supply button
58: air/water supply nozzle
59: suction button
60: treatment tool outlet port
61: through-hole
62: angle knob
64: treatment tool inlet port
66: elevator housing space
68: partition wall
68A: upper surface
74: illumination window
76: observation window
80: wire fixing member

120: link member
150: cantilever piece
150A: support piece
150B: stopped portion
160: stopper portion
162: groove portion
200: distal-end-cap detachment jig
202: body portion
204: inclined surface
206: inclined surface
208: plane
210: plane
212: plane
230: connection portion
230A: outer form
230B: bottom surface
230C: finger hole
250: connection portion
252: fixed arm
254: sliding arm
256: fulcrum portion
258: supporting surface
300: distal-end-cap detachment jig
400: distal-end-cap detachment jig
500: distal-end-cap detachment jig
510: body portion
510A: cylindrical part
510B: distal end part
530: connection portion
532: housing member
532A: space
532B: first opening
532C: bottom
532D: flange
532E: second opening
532F: space
Ax: major axis direction

What is claimed is:

1. A distal-end-cap detachment jig that is used to detach a distal end cap mounted on a distalend of a side-viewing scope, the distal end cap having an inner space which communicates with a cap opening allowing a passage of a treatment tool, the distal-end-cap detachment jig comprising:

a tubular-shaped housing member in which a space that houses the distal end cap is formed;

a first opening which is formed to the tubular-shaped housing member and through which the distal end cap is insertable;

a bottom which is formed to the tubular-shaped housing member and faces the first opening; and

a body portion which has a cylindrical shape and is configured to be insertable into the inner space, and of which one end in a longitudinal direction of the cylindrical shape is connected to the bottom of the tubular-shaped housing member in the space of the tubular-shaped housing member,

wherein the other end in the longitudinal direction of the cylindrical shape of the body portion has a tapered shape which is tapered toward a distal end side in a direction of insertion into the inner space of the distal end cap,

the body portion has a maximum width larger than a width of the inner space of the distal end cap, wherein the maximum width of the body portion and the width of the inner space is on a plane perpendicular to the direction of insertion, and

wherein the body portion is configured to enter the cap opening, deform the distal end cap and expand the inner space in the distal end cap.

2. The distal-end-cap detachment jig according to claim **1**, further comprising:

a connection portion that is connected to the body portion,

wherein the connection portion is connected at a position opposite to the direction of insertion of the body portion and is provided with a finger hole extending in a direction orthogonal to the direction of insertion.

3. The distal-end-cap detachment jig according to claim **2**, wherein the finger hole is a through-hole that penetrates the connection portion.

4. The distal-end-cap detachment jig according to claim **1**, further comprising:

a connection portion that is connected to the body portion,

wherein the connection portion includes a fixed arm, a sliding arm, and a fulcrum portion that connects one end of the fixed arm to one end of the sliding arm,

the body portion is connected to an end part of the sliding arm on a side opposite to the fulcrum portion, and

the body portion can be inserted into the inner space of the distal end cap with the fulcrum portion as a fulcrum.

5. The distal-end-cap detachment jig according to claim **4**, wherein a side of the fixed arm opposite to the fulcrum portion is formed by a curved surface that follows a shape of the distal end cap.

6. The distal-end-cap detachment jig according to claim **1**, wherein the space includes an escape space that houses the distal end cap deformed by the body portion.

7. The distal-end-cap detachment jig according to claim **1**, wherein the tubular-shaped housing member has a second opening that is opened toward a side orthogonal to the direction of insertion of the body portion.

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